KILLER POOL

Complete Design & Technical Specification

Version:	2.1 (Updated)
Document Date:	February 08, 2026
Project Type:	Multiplayer Android Pool Game
Target Platform:	Android (Unity Engine)
Game Variant:	Kenyan Cushion Pool - 13 Ball

Document Purpose: This comprehensive document specifies all design and technical requirements for developing Killer Pool, a unique multiplayer pool game variant popular in Kenya featuring a 13-ball cushion rack setup, sequential potting rules, carom scoring bonuses, and mathematical elimination mechanics.

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1. PROJECT OVERVIEW

Killer Pool is a multiplayer, turn-based Android pool game that replicates a unique Kenyan pool variant. Unlike traditional pool games, this variant features 13 balls arranged on the cushions (rails) rather than in a triangle, requires sequential potting (balls must be potted in numerical order 3→4→5...→15), and includes sophisticated mechanics such as carom scoring and mathematical elimination.

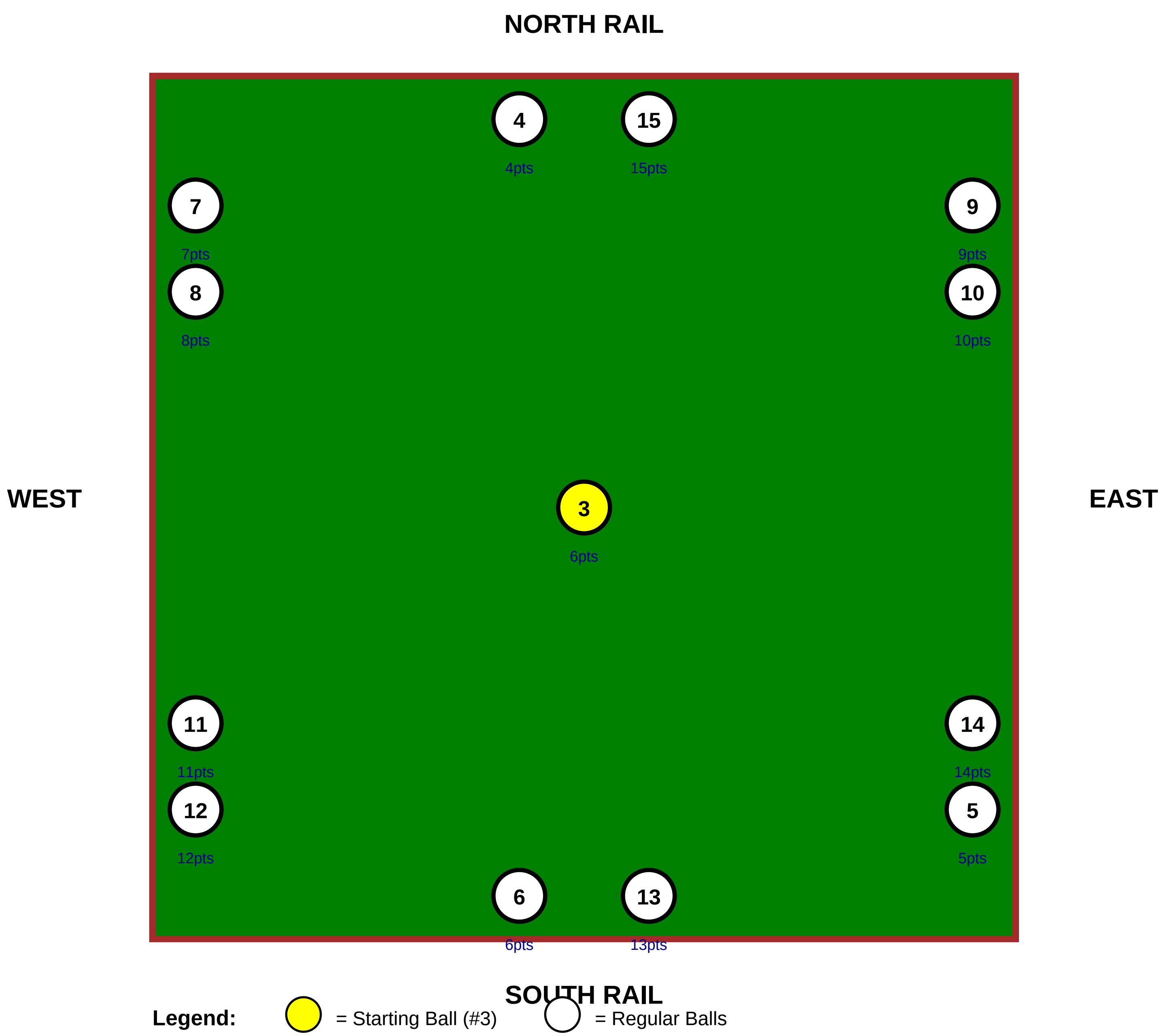
1.1 Key Distinguishing Features

Feature	Description
Ball Count	13 balls (numbered 3-15). Balls 1 & 2 not used
Ball Layout	Unique cushion arrangement with ball #3 at center
Play Style	Sequential: must pot balls in order 3→4→5...→15
Carom Scoring	Bonus points for potting extra balls on same shot
Bench Rule	Auto-eliminates players who cannot mathematically win
Economy	Virtual Kenyan Shilling (KSh) with tie-splitting support
Player Count	2-10 players with configurable lobby sizes
Payout System	10% service fee, net pool split among winners
AI Opponent	Available when no human opponents found online

2. VISUAL DIAGRAMS

2.1 Ball Layout Diagram - Cushion Rack Setup

The following diagram shows the precise placement of all 13 balls on the table. Unlike traditional pool where balls are racked in a triangle, Kenyan Cushion Pool features balls placed on the four rails (cushions) with one ball in the center.

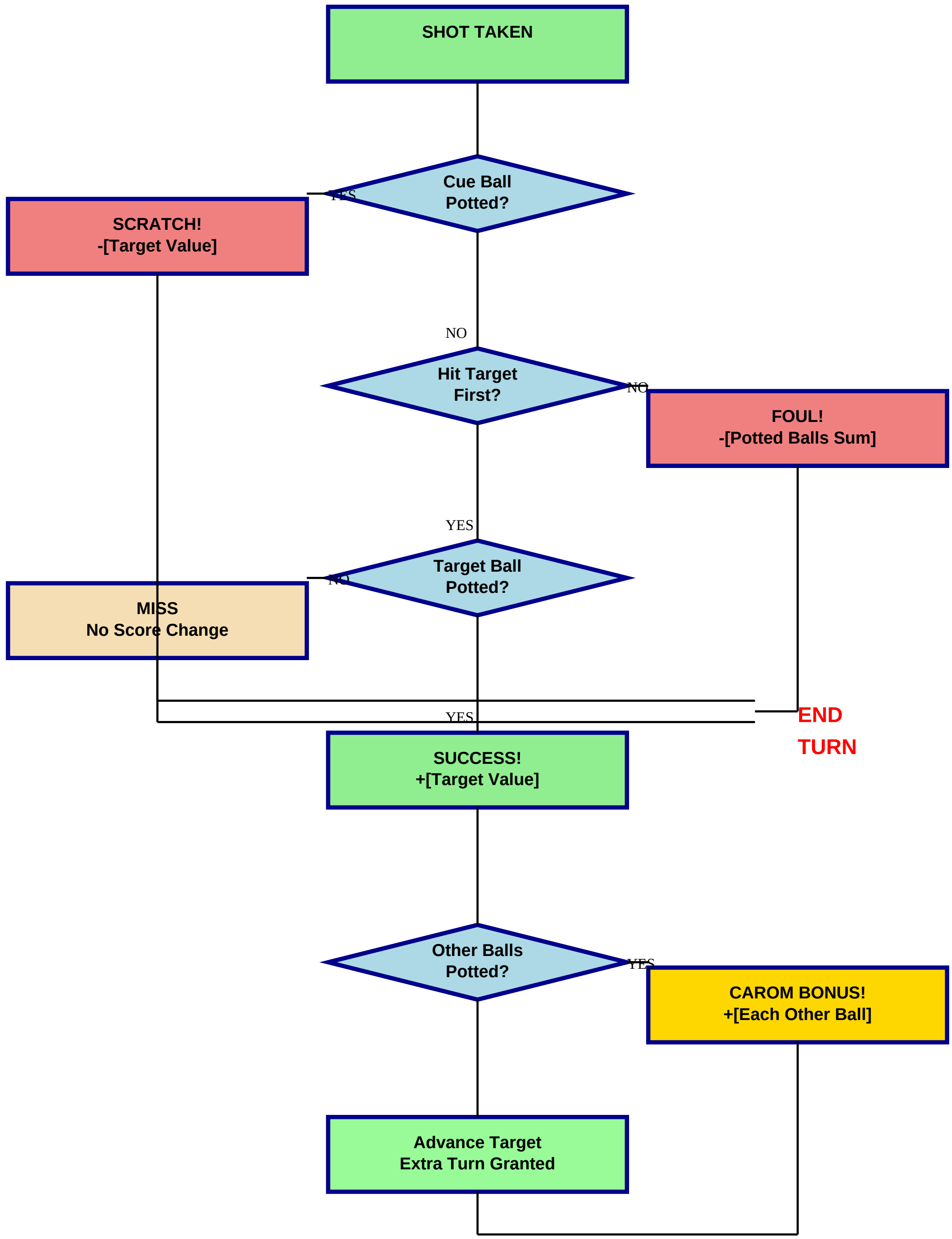


Ball Placement Summary:

Position	Ball Numbers	Total Balls
Center	#3 (6 pts)	1
North Rail	#4, #15	2
South Rail	#6, #13	2
West Rail	#7, #8, #11, #12	4
East Rail	#5, #9, #10, #14	4
TOTAL	**13 Balls**	**13**

2.2 Shot Resolution Flowchart

This flowchart illustrates the complete logic for resolving each shot, including foul detection, scoring, and the unique carom bonus rule.


SHOT TAKEN
Cue Ball Potted?
YES
SCRATCH! -[Target Value]
NO
Hit Target First?
NO
FOUL! -[Potted Balls Sum]
YES
Target Ball Potted?
NO
MISS No Score Change
YES
END TURN
SUCCESS! +[Target Value]
Other Balls Potted?
YES
CAROM BONUS! +[Each Other Ball]
Advance Target Extra Turn Granted

3. CORE GAME RULES & LOGIC

3.1 Objective & Win Condition

Players must pot balls in a strict numerical sequence: **3** → **4** → **5** → **6** → **7** → **8** → **9** → **10** → **11** → **12** → **13** → **14** → **15**

• **Total Balls:** 13 balls (numbered 3-15). Balls #1 and #2 are not used.

• **Game End:** When all 13 balls are potted.

• **Winner:** Player(s) with the highest cumulative score. **TIES SPLIT THE POT.**

3.2 The 'Bench' (Mathematical Elimination) Rule

A player is automatically 'benched' (eliminated from active play) if they cannot mathematically win or tie for the lead. This keeps the game dynamic and prevents unnecessary turns.

Bench Check Formula:

```
Potential Max = Current Score + Total Value of All Remaining Balls
```

Condition	Player Status	Action
Potential Max < Leader Score	BENCHED	Skip turn, spectate only
Potential Max = Leader Score	ACTIVE	Play normally (can tie)
Potential Max > Leader Score	ACTIVE	Play normally (can win)

• **Re-Activation:** Automatic if player's potential maximum becomes ≥ leader's score due to other players' actions.

3.3 Turn Structure & Scoring (With Carom Rule)

Each player's turn consists of ONE shot at their current target ball. The unique **carom rule** allows players to earn bonus points by potting additional balls beyond their target.

Outcome	Action	Result
SUCCESS + CAROM	Cue ball hits target first, pots target AND other ball(s)	Score +[target] +[each other ball] Target advances Extra turn granted
SUCCESS (Clean)	Only target ball potted	Score +[target value] Target advances Extra turn granted
FOUL - Wrong Contact	Cue ball hits wrong ball first, pots any balls	Score -[sum of potted balls] Target unchanged Turn ends
FOUL - Scratch	Cue ball is potted	Score -[target value] Target unchanged Turn ends
MISS	No balls potted	No score change Turn ends

Carom Scoring Example:

Player targeting ball #7 successfully pots #7, and also pots #10 and #15 in the same shot.

• Score: +7 (target) +10 (carom) +15 (carom) = **+32 points total**

• Target advances from #7 to #8

• Player receives an extra turn

3.4 Ball Values (Marks)

Each ball has a specific point value. Note the special case for ball #3.

Ball Number	Point Value	Notes
#3	6 points	Starting ball - special double value
#4 through #15	Equal to number	#4=4 pts, #5=5 pts, ... #15=15 pts

Total Possible Points Calculation:

• Ball #3: 6 points

• Balls #4-15: 4+5+6+7+8+9+10+11+12+13+14+15 = 99 points

• **Total: 6 + 99 = 105 points**

3.5 Multiplayer & Session Flow

The game supports 2 to 10 players with configurable lobby options.

• **Player Count:** 2 to 10 players

• **Lobby Options:** 2 players, 3-5 players, 5-7 players, 7-10 players

• **Turn Order:** Fixed random order assigned before game starts

• **Online Matchmaking:** If no human opponent found within 30 seconds, player is prompted to either WAIT or PLAY VS COMPUTER

• **Between Rounds:** Winner(s) receive payout. Players choose CONTINUE (reset to ball #3) or QUIT

4. VIRTUAL ECONOMY & PAYOUT SYSTEM

The game features a virtual Kenyan Shilling (KSh) economy that supports both single winners and tie scenarios. All transactions use virtual currency with no real-world monetary value.

Component	Value/Rule	Description
Starting Balance	100,000 KSh	Every new player receives this virtual currency
Stake Per Game	5,000 KSh	Fixed amount deducted when game starts (configurable)
Prize Pool	Stake × Players	Total pool before service fee
Service Fee	10% of pool	Platform fee deducted from prize pool
Net Pool	Total × 0.90	Amount available for winners
Single Winner	100% of Net Pool	Winner receives entire net pool
Tie Payout	Net Pool ÷ Winners	Net pool split equally among tied winners

Example Payout Calculations:

Scenario 1: Single Winner (4 players, 5,000 KSh stake)

• Total Pool: 4 × 5,000 = 20,000 KSh

• Service Fee: 20,000 × 0.10 = 2,000 KSh

• Net Pool: 20,000 - 2,000 = 18,000 KSh

• Winner receives: 18,000 KSh

Scenario 2: Two-Way Tie (4 players, 5,000 KSh stake)

• Total Pool: 4 × 5,000 = 20,000 KSh

• Service Fee: 20,000 × 0.10 = 2,000 KSh

• Net Pool: 20,000 - 2,000 = 18,000 KSh

• Each winner receives: 18,000 ÷ 2 = 9,000 KSh

5. TECHNICAL IMPLEMENTATION

5.1 Required Game State Variables

Core Game State:

• **List<Player> allPlayers** - All players in the game

• **List<Ball> ballsOnTable** - 13 balls with current state

• **int currentTurnIndex** - Index of player whose turn it is

• **int prizePool** - Current game's prize pool in KSh

• **GamePhase** - Current phase (Lobby, InProgress, RoundOver)

Ball Class Properties:

• **int ballNumber** - Number on the ball (3-15)

• **int pointValue** - Points this ball is worth

• **bool isPotted** - Whether ball has been potted

• **Vector3 position** - Current 3D position on table

Player Class Properties:

• **string id** - Unique player identifier

• **int score** - Current cumulative score

• **Ball currentTargetBall** - Reference to target ball object

• **bool isBenched** - Whether player is mathematically eliminated

• **float balance** - Virtual currency balance in KSh

• **int position** - Turn order position (1 to N)

• **bool isAI** - Whether player is computer-controlled opponent

5.2 UI/UX Requirements

In-Game HUD (Heads-Up Display):

• Large Target Display: Show 'TARGET: #3 (6 pts)' prominently

• Score Display: Current player score and leaderboard

• Benched Overlay: 'BENCHED' indicator when player is eliminated

• Carom Notification: Pop-up showing 'Carom! +4 +7' for bonus balls

• Leaderboard: Real-time ranking of all players

• AI Indicator: Show (AI) next to computer opponent's name

Matchmaking & Opponent Selection UI:

• 'Waiting for Opponent...' screen with 30-second timer

• 'Play vs Computer' button

• 'Keep Waiting' button

• Estimated wait time display

Round End Screen:

Must properly handle and display tie scenarios:

• Single Winner: 'Victory! You won 18,000 KSh'

• Tie: 'Tie Game! Net Pool: 18,000 KSh. Split 2 ways. You win 9,000 KSh'

• Option buttons: CONTINUE or QUIT

5.3 Shot Resolution Algorithm (Pseudo-Code)

The shot resolution algorithm is the heart of the game logic. It processes each shot and determines scoring, fouls, and turn advancement.

```
ResolveShot(Player player, Ball firstBallHit, List<Ball> allPottedBalls, bool isCueBallPotted) { // 1. Check for Scratch Foul (cue ball potted) if (isCueBallPotted) { player.score -= player.currentTargetBall.value; endTurn(); return; } // 2. Check for Wrong First Contact Foul if (firstBallHit != player.currentTargetBall) { int penalty = 0; foreach (Ball b in allPottedBalls) { penalty += b.value; } player.score -= penalty; endTurn(); return; } // 3. SUCCESS - Player hit correct target first bool wasTargetPotted = allPottedBalls.Contains(player.currentTargetBall); int bonusPoints = 0; if (wasTargetPotted) { // Award points for target player.score += player.currentTargetBall.value; // Award bonus for any OTHER balls potted (carom) foreach (Ball b in allPottedBalls) { if (b != player.currentTargetBall) { bonusPoints += b.value; } } player.score += bonusPoints; // Advance target and grant extra turn advanceTargetBall(player); grantExtraTurn(); } else { // Edge case: Hit target first but didn't pot it (miss) endTurn(); } }
```

6. AI CODING ASSISTANCE PROMPTS

The following prompts are designed to be used sequentially with AI coding assistants (ChatGPT, Claude, GitHub Copilot, etc.) to generate the core game code.

#	Focus Area	Prompt
1	Data Structures	Create C# classes for Unity for the game "Killer Pool": Ball (with number, pointValue, isPotted) and Player. For the Game
2	Physics Detection	In a Unity pool game using 3D physics, how do I detect and log: 1) The first ball the cue ball hits, and 2) All balls potted o
3	Game Logic	Write the main GameLogicController C# script. Include: ResolveShot function implementing carom and foul rules, Check
4	Economy System	Create the EconomyManager C# class with static methods. Include ProcessPayout which takes the total stake pool and
5	Matchmaking AI	Create a MatchmakingManager C# class that: 1) Searches for online opponents for 30 seconds 2) If none found, display

7. DEVELOPMENT PHASES & ROADMAP

Phase	Key Activities	Deliverables
1: Pre-Dev Setup	• Create visual mockups • Set up Unity & Android SDK • Configure Firebase • Choose networking (Photon/Mirror) • Define art style	• Development environment ready • Project structure created • Asset pipeline established
2: Core Mechanics	• Implement 3D pool physics • Build 13-ball cushion setup • Create turn-based game loop • Implement carom scoring • Build bench rule logic • Add basic AI opponent	• Playable single-player prototype • All core rules functional • AI opponent working
3: UI/UX	• Build all screens and menus • Implement HUD elements • Add visual feedback systems • Create leaderboard UI • Build economy interface • Design matchmaking UI	• Complete UI system • Polished user experience • Matchmaking interface
4: Multiplayer	• Implement networking layer • Synchronize game state • Build matchmaking system • Add authentication • Server-side validation • Integrate AI fallback	• Working multiplayer • Secure client-server system • Matchmaking with AI option
5: Testing	• Test bench rule edge cases • Validate carom scoring • Test tie scenarios • Performance optimization • Network latency testing • AI difficulty balancing	• Bug-free game • Optimized performance • Balanced gameplay
6: Polish & Launch	• Add sound effects & music • Implement achievements • Social features • Create tutorial • Analytics tracking • Store submission prep	• Polished final product • Google Play ready • Launch materials prepared

8. APPENDICES

8.1 Complete Ball Values Reference

Ball #	Points	Ball #	Points	Ball #	Points
3	6	8	8	13	13
4	4	9	9	14	14
5	5	10	10	15	15
6	6	11	11		
7	7	12	12		

8.2 Bench Rule Calculation Examples

Scenario	Details	Result
Player Stays Active	Player A: 40 pts Player B: 65 pts (Leader) Remaining: #12,#13,#14,#15 = 54 pts Potential Max: 40 + 54 = 94	94 > 65 Player A ACTIVE
Player Gets Benched	Player A: 40 pts Player B: 90 pts (Leader) Remaining: #14,#15 = 29 pts Potential Max: 40 + 29 = 69	69 < 90 Player A BENCHED
Tie Protection	Player C: 76 pts Player D: 90 pts (Leader) Remaining: #14 = 14 pts Potential Max: 76 + 14 = 90	90 = 90 Player C ACTIVE (can tie)

8.3 Recommended Technology Stack

Component	Recommended Solution	Alternative Options
Game Engine	Unity 2022.3 LTS	Unity 2023.x, Unreal Engine
Language	C#	N/A (Unity standard)
Networking	Photon PUN 2	Mirror, Netcode for GameObjects
Backend/Database	Firebase (Auth, Firestore)	PlayFab, AWS GameLift
Physics	Unity PhysX + Pool Asset	Custom implementation
Analytics	Unity Analytics + Firebase	GameAnalytics
Version Control	Git + GitHub/GitLab	Plastic SCM
IDE	Visual Studio 2022	Rider, VS Code

8.4 Document Version History

Version	Date	Changes
2.1	February 08, 2026	Added AI opponent fallback in matchmaking Extended Player class with isAI property Updated UI/UX for opponent selection Added Prompt 5 for AI matchmaking
2.0	January 2026	Complete specification created Added all diagrams and examples Finalized technical requirements

END OF DOCUMENT

This is a living specification and should be updated as development progresses.

For questions or clarifications, maintain a changelog of decisions and rationale.